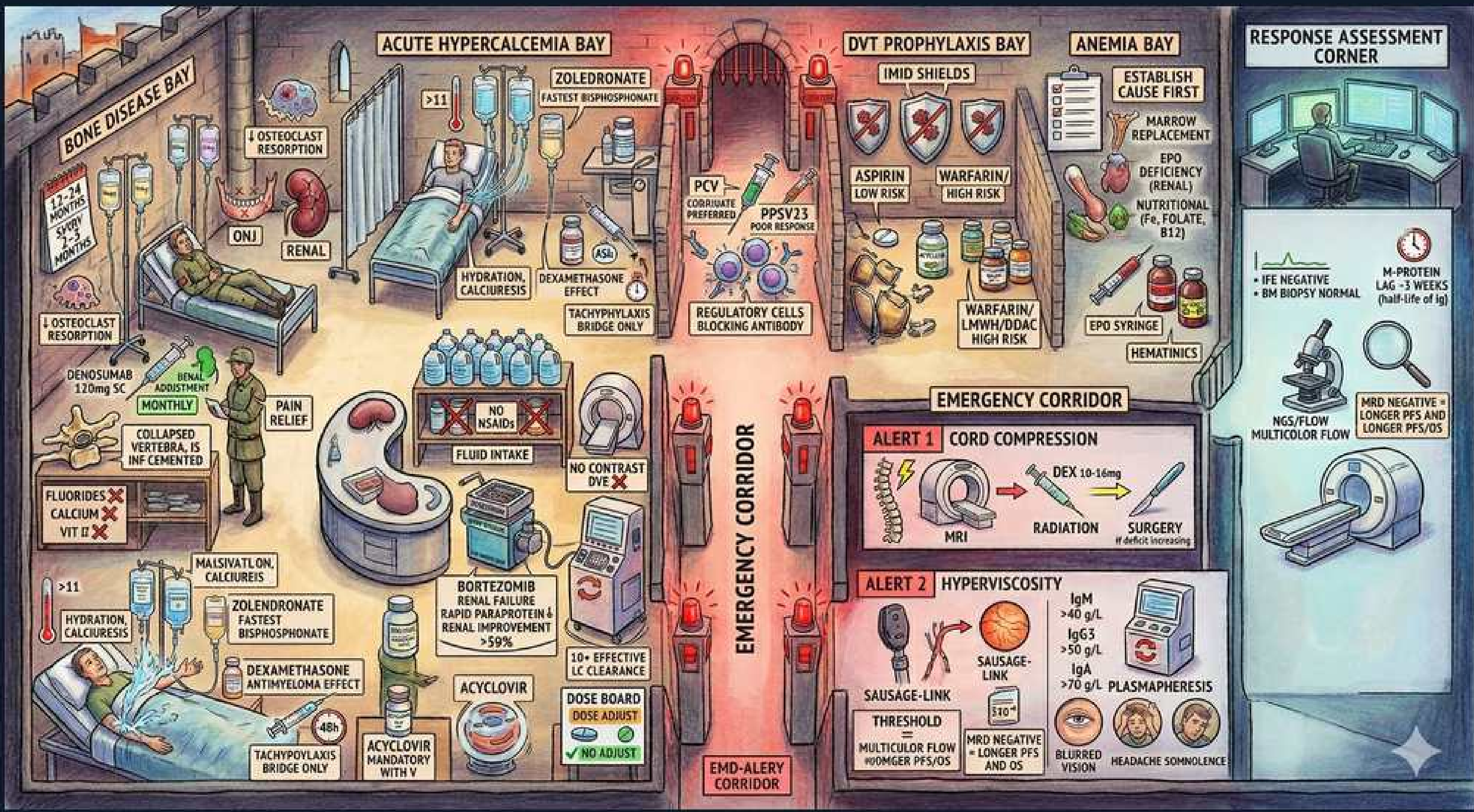
Plasma Cell — Myeloma Emergencies & Complications



1. Bone disease

WHAT YOU SEE

The BONE DISEASE BAY (top-left) showing osteoclast resorption and a collapsed vertebra — give a bone agent (zoledronate or denosumab) to every patient with myeloma bone disease.

WHAT IT ENCODES

Myeloma drives RANKL-mediated osteoclast activation and DKK1/sclerostin-mediated osteoblast suppression → pure lytic lesions, fractures, and hypercalcemia. Give a bone-modifying agent (zoledronate 4 mg IV q3–4 weeks, or denosumab) to ALL patients with bone disease to reduce skeletal-related events. Supplement calcium + vitamin D and screen dental health before bisphosphonate/denosumab (osteonecrosis of the jaw risk).

BOARD TRAP

WRONG to order a technetium bone scan to find or follow myeloma bone lesions — it is falsely negative because osteoblasts are suppressed (no reactive uptake). Discriminator: use whole-body low-dose CT, whole-body MRI, or FDG-PET/CT; the bone scan is the classic distractor answer.

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2. Denosumab option

WHAT YOU SEE

DENOSUMAB 120 mg monthly syringe in the bone bay — an anti-RANKL antibody that is NOT renally cleared, the preferred bone agent when kidneys are impaired.

WHAT IT ENCODES

Denosumab is a fully human anti-RANKL monoclonal antibody (120 mg SC monthly for bone disease) that blocks osteoclast activation. It is NOT renally cleared, making it the preferred bone agent in renal impairment where zoledronate is contraindicated/nephrotoxic. Monitor for hypocalcemia (ensure adequate Ca/vitamin D) and osteonecrosis of the jaw; avoid abrupt discontinuation (rebound bone loss).

BOARD TRAP

WRONG to keep using zoledronate in a myeloma patient with CrCl <30–35 mL/min. Switch to denosumab (no renal dosing) but pre-empt hypocalcemia. Discriminator: renal function picks the agent — denosumab for renal impairment, bisphosphonate when renal function is adequate; both require dental clearance for ONJ.

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3. RT for pain/cord

WHAT YOU SEE

RADIATION / vertebroplasty panel over the collapsed vertebra — focal low-dose RT gives rapid pain relief and treats plasmacytoma/cord compression in this radiosensitive tumor.

WHAT IT ENCODES

Low-dose external-beam radiotherapy (typically 8 Gy single fraction or 20–30 Gy fractionated) gives rapid relief of focal uncontrolled bone pain, treats solitary plasmacytoma, and is part of cord-compression management. Vertebroplasty/kyphoplasty stabilizes painful compression fractures. Myeloma is highly radiosensitive.

BOARD TRAP

WRONG to irradiate large marrow-containing fields routinely — excessive RT damages marrow reserve and compromises ability to deliver systemic chemo/transplant. Discriminator: use focal low-dose RT for localized pain/cord/plasmacytoma; systemic therapy (not broad RT) treats the disease as a whole.

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4. Hypercalcemia Rx

WHAT YOU SEE

The ACUTE HYPERCALCEMIA BAY with IV saline bags and zoledronate — step 1 is aggressive IV saline, step 2 an IV bisphosphonate.

WHAT IT ENCODES

Hypercalcemia of malignancy (corrected Ca >11; severe >14 mg/dL or symptomatic): step 1 is aggressive IV isotonic saline (200–300 mL/hr, correcting the near-universal volume depletion); step 2 is an IV bisphosphonate (zoledronate 4 mg, onset 2–4 days, peak ~4–7 days); calcitonin (4 IU/kg q12h) is a rapid bridge for the first 24–48 hr. Denosumab is an option in renal failure or refractory cases.

BOARD TRAP

WRONG to start a thiazide diuretic — thiazides INCREASE distal calcium reabsorption and worsen hypercalcemia. Loop diuretics are not first-line either and only have a role AFTER full volume repletion if there is volume overload. Discriminator: saline first, never lead with diuretics; thiazide is always the wrong answer in hypercalcemia.

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5. Calcitonin bridge

WHAT YOU SEE

CALCITONIN vial tagged 'tachyphylaxis / bridge only' — it drops calcium fast but receptors downregulate in 48-72h, so it is only a short bridge.

WHAT IT ENCODES

Calcitonin lowers calcium quickly (within hours) by inhibiting osteoclastic resorption and increasing renal Ca excretion, but tachyphylaxis develops in 48–72 hr (downregulated receptors), so it serves only as a short bridge while saline + bisphosphonate take effect. Modest effect (drops Ca ~1–2 mg/dL).

BOARD TRAP

WRONG to rely on calcitonin as definitive hypercalcemia therapy. Its effect is small and transient (tachyphylaxis), so it must be paired with the slower, durable bisphosphonate/denosumab plus saline. Discriminator: calcitonin = fast but fleeting bridge; the bisphosphonate provides the sustained calcium control.

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6. Steroids for HyperCa

WHAT YOU SEE

DEXAMETHASONE in the hypercalcemia bay — steroids lower calcium in myeloma (tumor-lytic) and in calcitriol-driven granulomatous/lymphoma hypercalcemia.

WHAT IT ENCODES

Corticosteroids (e.g., dexamethasone) lower calcium in myeloma by direct anti-plasma-cell (tumor-lytic) effect and by reducing intestinal calcium absorption. They are also the targeted treatment for hypercalcemia from lymphoma and granulomatous disease (sarcoid/TB), where they cut 1-alpha-hydroxylase-driven calcitriol production by macrophages.

BOARD TRAP

WRONG to use steroids as first-line for parathyroid (primary HPT) or solid-tumor PTHrP hypercalcemia — they do not work there. Discriminator: steroids lower calcium specifically in calcitriol-mediated (granulomatous/lymphoma) and tumor-lytic (myeloma) hypercalcemia, not in PTH/PTHrP-driven hypercalcemia.

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7. Bortezomib in renal failure

WHAT YOU SEE

BORTEZOMIB panel marked 'no dose reduction in renal failure' — the proteasome inhibitor needs no renal adjustment and rapidly clears light chains, the backbone for myeloma kidney.

WHAT IT ENCODES

Bortezomib (a proteasome inhibitor) requires NO renal dose adjustment, is not dialyzed off significantly, and rapidly lowers the light-chain burden — making bortezomib-based regimens (e.g., VCd or VRd) the preferred initial therapy to reverse cast nephropathy/AKI in newly diagnosed myeloma with renal failure. Pair with aggressive hydration and avoid nephrotoxins/contrast.

BOARD TRAP

WRONG to give full-dose lenalidomide in renal impairment. Lenalidomide IS renally cleared and requires dose reduction by CrCl (and dialysis dosing). Discriminator: bortezomib needs no renal adjustment and is the renal-failure backbone; lenalidomide must be dose-reduced. Also watch bortezomib's peripheral neuropathy — give it weekly and subcutaneously to limit it.

8. HZV prophylaxis

WHAT YOU SEE

An ACYCLOVIR capsule paired with bortezomib/dara — proteasome inhibitors and anti-CD38 reactivate varicella-zoster, so give antiviral prophylaxis throughout.

WHAT IT ENCODES

Proteasome inhibitors (bortezomib, carfilzomib, ixazomib) and anti-CD38 antibodies (daratumumab) markedly increase varicella-zoster reactivation risk → give acyclovir (or valacyclovir) prophylaxis throughout treatment. Anti-CD38 also necessitates blood-bank notification (interferes with indirect Coombs/crossmatch — use DTT-treated cells) and can cause infusion reactions (premedicate).

BOARD TRAP

WRONG to omit antiviral prophylaxis when starting bortezomib or daratumumab. Untreated, herpes zoster reactivation (including disseminated/ophthalmic) is common and preventable. Discriminator: every proteasome inhibitor or anti-CD38 patient gets acyclovir/valacyclovir VZV prophylaxis.

9. DVT prophylaxis

WHAT YOU SEE

The DVT PROPHYLAXIS BAY: IMiD shields with aspirin vs anticoagulant options — IMiDs + dex raise clot risk, so stratify (aspirin low-risk, LMWH/DOAC high-risk).

WHAT IT ENCODES

Immunomodulatory drugs (thalidomide, lenalidomide, pomalidomide) — especially combined with dexamethasone and/or erythropoietin — substantially raise venous thromboembolism risk. Stratify: low-risk patients → aspirin 81–325 mg; high-risk (prior VTE, immobility, central line, high-dose dex, EPO use) → prophylactic LMWH or a DOAC.

BOARD TRAP

WRONG to start an IMiD without VTE prophylaxis or to default everyone to aspirin. High-risk patients need LMWH/DOAC, not aspirin. Discriminator: IMiD + dex (or EPO) mandates risk-stratified thromboprophylaxis; high thrombotic risk overrides the convenience of aspirin.

10. Infection / vaccines

WHAT YOU SEE

PCV/PPSV23 pneumococcal vaccines + immunoglobulin replacement in the anemia/infection bay — immunoparesis makes encapsulated-organism infection a leading killer, so vaccinate and consider IVIG.

WHAT IT ENCODES

Infection is a LEADING cause of death in myeloma due to functional hypogammaglobulinemia (immunoparesis), neutropenia, and treatment immunosuppression — encapsulated organisms (S. pneumoniae, H. influenzae) and gram-negatives predominate. Vaccinate (pneumococcal PCV then PPSV23, influenza, COVID; avoid live vaccines), consider IVIG for recurrent severe infections with low IgG, and PCP prophylaxis with high-dose steroids.

BOARD TRAP

WRONG to interpret a high total serum protein as evidence of intact immunity. The M-protein is one useless monoclonal antibody while normal polyclonal Ig is suppressed (immunoparesis), so patients are functionally immunodeficient despite hypergammaglobulinemia. Discriminator: check uninvolved/quantitative Ig levels — high total protein with low normal Ig = vaccinate, prophylax, and treat infection aggressively.

11. Anemia / EPO

WHAT YOU SEE

The ANEMIA BAY with erythropoietin and an iron/B12/folate workup — the normocytic 'A' of CRAB is multifactorial (marrow infiltration + IL-6/hepcidin + renal EPO loss).

WHAT IT ENCODES

Normocytic anemia (the 'A' of CRAB) is multifactorial: marrow plasma-cell infiltration, IL-6/hepcidin-driven anemia of inflammation, and renal EPO deficiency from cast nephropathy. Rouleaux formation on smear (stacked RBCs) and a high ESR reflect the paraprotein. Work up reversible causes (iron, B12, folate), treat the myeloma, and consider EPO for symptomatic anemia.

BOARD TRAP

WRONG to reflexively transfuse or give iron without a workup, or to ignore that effective anti-myeloma therapy is the best anemia treatment. EPO also adds VTE risk on top of IMiDs. Discriminator: anemia is multifactorial — treat the clone, correct true deficiencies, and weigh EPO's thrombosis risk before starting it.

12. Cord compression ALERT

WHAT YOU SEE

ALERT 1: CORD COMPRESSION panel — MRI + dexamethasone + RT/surgery — a true emergency: give high-dose dex on clinical suspicion BEFORE the confirming MRI.

WHAT IT ENCODES

Malignant spinal cord compression (back pain, then weakness, sensory level, bowel/bladder dysfunction) is a TRUE emergency. Give high-dose dexamethasone IMMEDIATELY on clinical suspicion, obtain urgent whole-spine MRI, then definitive RT (myeloma is radiosensitive) or surgical decompression if unstable/structural. Outcome depends on neurologic status at treatment — early action preserves function.

BOARD TRAP

WRONG to delay steroids until MRI confirms compression. If the exam suggests cord compression, give dexamethasone first — do not wait for imaging. Discriminator: steroids are time-critical and started on clinical suspicion; the MRI confirms but should not gate the first dose.

13. Hyperviscosity ALERT

WHAT YOU SEE

ALERT 2: HYPERVISCOSITY panel showing 'sausage-link' dilated retinal veins and a plasmapheresis machine — emergent plasma exchange removes the paraprotein (classically large IgM/Waldenström).

WHAT IT ENCODES

Hyperviscosity syndrome = mucosal bleeding (epistaxis, gingival), visual changes, headache/confusion, and the classic 'sausage-link'/dilated tortuous retinal veins on fundoscopy. Emergent treatment is plasmapheresis (plasma exchange) to rapidly remove the paraprotein, followed by chemotherapy for the underlying clone. Most common with large IgM (Waldenström) > IgA > IgG.

BOARD TRAP

WRONG to wait for a specific serum viscosity number before treating, or to assume myeloma when the M-protein is IgM. Treat the clinical syndrome with plasmapheresis regardless of the exact viscosity value, and remember IgM-driven hyperviscosity = Waldenström macroglobulinemia, not classic myeloma. Discriminator: clinical hyperviscosity (retina + bleeding + neuro) → plasmapheresis now; IgM points to Waldenström.

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14. Response assessment

WHAT YOU SEE

The RESPONSE ASSESSMENT CORNER with an IFE-negative readout and MRD flow scanner — deeper responses (IFE-negative > SPEP-normal; MRD-negative deepest) predict longer survival.

WHAT IT ENCODES

Response depth predicts outcome: stringent complete response and especially MRD-negativity (by next-gen flow or sequencing, sensitivity 10^-5 to 10^-6) correlate with longer PFS and OS. Immunofixation (IFE)-negative complete response is deeper than SPEP normalization. Monitor with SPEP/UPEP, serum free light chains, and IFE; use the involved FLC for oligosecretory disease.

BOARD TRAP

WRONG to call a patient in complete response based on a normal SPEP alone. SPEP can be normal while immunofixation still shows residual M-protein, and MRD testing detects disease below IFE. Discriminator: deeper response definitions (IFE-negative > SPEP-normal; MRD-negative > IFE-negative) — and serum FLC is required to track light-chain/oligosecretory disease.
